

Video Analytics with Deep Learning: Methods and
Applications

Intelligent Video Analytics with Deep Learning

December & Company Inc.

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Intelligent Video Analytics with Deep Learning

- (Subtitle) “Video Analytics with Deep Learning: Methods and Applications”

Goals of This Session

- Video Analytics with Deep Learning: Methods and Applications



- There is no prerequisite related with programming skills for this session

Goals of This Session

- Video Analytics with Deep Learning: **Methods and Applications**



- There is no prerequisite related with programming skills for this session.
- Make it possible for you to analyze your own video data

What is Video Analytics?

- Video Analytics is really the study of video data
- The amount of video data in our world has exploded to a ridiculous degree
 - 82% of all Internet traffic would be video by 2022 (CISCO, 2018)
 - The majority of bits flying around the Internet are actually video



Why is it Difficult to Analyze Video Data?

- However, there's a problem with video data that it's really hard to understand
 - Video has multiple modalities with the usage of audio, visual and (possibly) textual data
 - A few of Problems: Semantic Gap, Deformation, Illumination, Occlusion, Intra-class Variation, Cocktail Party Effect, etc

Why is it Difficult to Analyze Video Data?

- Challenges: Deformation



Why is it Difficult to Analyze Video Data?

- Challenges: Occlusion



Why is it Difficult to Analyze Video Data?

- Challenges: Intra-class Variation



Why is it Difficult to Analyze Video Data?

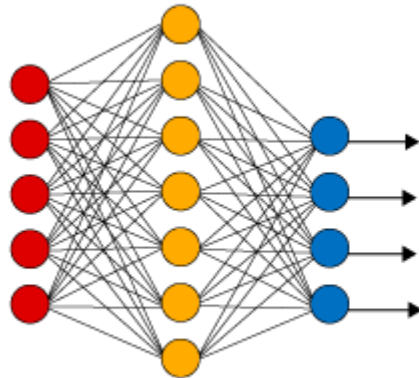
- Challenges: Cocktail Party Effect



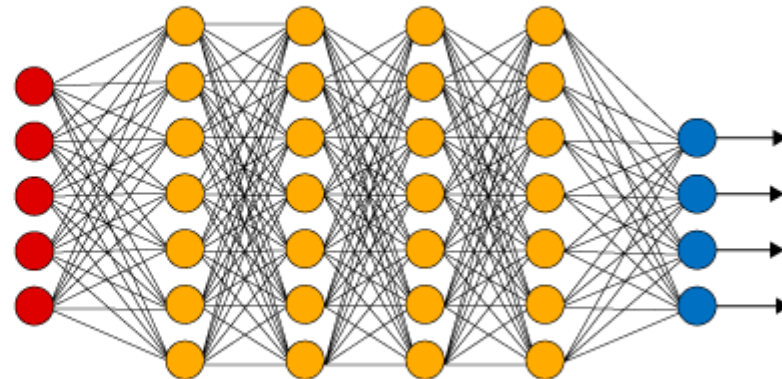
Why is Video Analytics Recently Becoming a Big Player?

- Deep Learning has achieved very promising results in a wide range of areas including **Computer Vision**, **Speech Recognition** and **Natural Language Processing**

Simple Neural Network



Deep Learning Neural Network



● Input Layer

● Hidden Layer

● Output Layer

Key Market Segments of Video Analytics

- Video Analytics Market was valued at \$2,745 million in 2016, and is estimated to reach \$13,381 million by 2023 (Allied Market Research, 2017)

Video Analytics Market Key Segments	
By Application	- Activity and Event Recognition - Audio and Speech Processing - Scene Analysis and Interpretation - Visual Feature Extraction - Question Answering
Industry Vertical	- Entertainment - Healthcare - Home Automation - Transportation - Retail - Manufacturing - Energy & Utilities - Critical Infrastructure

<https://datafloq.com/read/what-is-video-analytics-why-becoming-big-player/2312>

<https://www.alliedmarketresearch.com/video-analytics-market>

Intelligent Video Analytics with Deep Learning

- Welcome to “Intelligent Video Analytics with Deep Learning”
- We mainly focus on **1. social science research and practice applying video analytics** and **2. learn how to understand and analyze quantitative video data** based on deep learning

Three Modules of This Session

- Module 1. Video Analytics and Social Science Research
- Module 2. Underlying Mechanisms of Deep Learning
- Module 3. Detecting Labels in Your Own Video Data

Thank you ☺

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